CSE 3200: System Development Project

# Horizons: A Serious Game for Autism Spectrum Disorder Pre-Screening

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**April, 2026**

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**Authors**

# Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition marked by persistent differences in social communication and by restricted, repetitive patterns of behaviour. Validated direct-observation instruments such as ADOS-2 remain inaccessible to many families because they require trained clinicians, controlled environments and considerable administration time. This project presents *Horizons*, a browser-based serious game that collects structured behavioural signals from children aged 3 to 10 across six interconnected gameplay chapters. The chapters address social communication, restricted and repetitive behaviours, sensory processing and pretend play, drawing from the domain structure of ADOS-2, DSM-5 and M-CHAT-R/F. For every task the game records response time, selection accuracy, attempt count and pointer movement trajectory, which feed a penalty-based scoring engine that computes weighted domain scores and a four-level combined risk classification.

The system runs on Next.js 15 with the Bun runtime, PostgreSQL hosted on the Neon serverless platform via `postgres.js`, Better Auth for role-based researcher authentication, Tone.js for real-time Web Audio sound synthesis and Tailwind CSS v4 with ShadCN UI for the interface. No image or pre-recorded audio files are shipped; visual content is rendered entirely from emoji and CSS and every game sound is synthesised at runtime by Tone.js. A public-facing platform provides landing, about, contact and survey pages. At session completion the scoring engine generates an HMAC-SHA256 signed caregiver report link that can be forwarded directly to a healthcare provider. An authenticated researcher dashboard, secured by Better Auth, delivers per-session domain charts, pointer heatmaps, red-flag summaries and JSON or CSV data exports. An automated test suite of 312 tests covers scoring logic, database queries, API routes and the full scoring pipeline.

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# CHAPTER I

# Introduction

## 1.1 Introduction

Approximately one out of 36 children in the USA have ASD [1], while roughly 1–2% of children globally suffer from this condition [2]. These presentations are heterogeneous on their social-communicative spectrum and display varying degrees of restricted/repetitive behaviors; hence, there is no single route to screening that could satisfy all parents or clinical scenarios. Direct observation tools, especially the Autism Diagnostic Observation Schedule (ADOS-2) [3], generate reliable behavioral data, but they are dependent on qualified professionals, standard assessment locations and extensive evaluation periods. This makes them impractical for many parents who do not have sufficient financial resources.

Serious games represent an alternative solution to this problem. With a well-designed game, validated behavioral activities can be incorporated into typical gameplay experiences, allowing structured observations to take place without the rigors of formal assessment sessions. Irani et al. [4] showed that a mobile emotion recognition game reached 93% screening accuracy using support vector machine classification, confirming that gameplay response data can carry clinically meaningful signal.

Horizons was built around that premise. The system is a six-chapter browser-based serious game that collects multi-domain behavioural data from children aged 3 to 10, running on any modern-browser device and requiring no installation, dedicated hardware or clinician presence beyond basic supervision.

## 1.2 Background and Problem Statement

Early ASD screening tools differ substantially in validity, accessibility and ecological validity. Caregiver questionnaires such as M-CHAT-R/F [5] are practical and widely used, but they depend on accurate parental recall and are therefore vulnerable to observer bias. Direct-observation instruments such as ADOS-2 are more accurate but expensive and time-consuming to administer. Tablet and browser-based serious games have emerged as a middle ground; however, most existing implementations run on proprietary platforms, require application installation, cover only a narrow age range or do not collect the timestamped multi-domain behavioural data that rigorous screening requires [6].

The core problem this project addresses is as follows: how can a production-standard, browser-delivered serious game simultaneously provide an engaging experience for children aged 3 to 10 and collect clinically meaningful, multi-domain ASD screening signals without requiring specialised equipment or professional administration?

## 1.3 Objectives

The primary objectives of this project are:

1. To design and implement a browser-based serious game spanning six chapters that together cover the four ASD indicator domains defined by ADOS-2, DSM-5 and M-CHAT-R/F.

2. To develop a penalty-based multi-domain scoring engine grounded in three peer-reviewed research frameworks: ADOS-2 domain weighting, DTT-based game scoring and the EmoGalaxy task accuracy formula.

3. To implement automatic red-flag detection with multiplicative severity weighting for clinically significant behavioural patterns.

4. To build a researcher dashboard with Better Auth role-based authentication, per-session analytics, pointer heatmaps and JSON or CSV exports.

5. To generate HMAC-SHA256 signed caregiver report links at session completion that a healthcare provider can access without researcher involvement.

## 1.4 Scope

Horizons encompasses the full child-facing gameplay flow across six chapters, the scoring and risk-classification engine, public platform pages, the authenticated researcher dashboard and the HMAC-signed caregiver report generation. The principal technologies employed are:

- **Next.js 15 (App Router):** Server components, file-based routing and API route handlers for clean separation of client and server logic.
- **Bun runtime:** Built-in test runner, native package resolution and the `bun` package manager.
- **PostgreSQL via Neon:** Serverless PostgreSQL with the `postgres.js` driver; schema migration runs automatically on every server start.
- **Better Auth:** Open-source email and password authentication with researcher and admin roles, HttpOnly session cookies and in-memory rate limiting.
- **Tailwind CSS v4 with `@theme` tokens:** Design-token-driven theming without raw CSS files beyond `globals.css`.
- **ShadCN UI:** Radix-based accessible component library.
- **Framer Motion:** Physics-based animation with sensory-intensity gating.
- **Tone.js:** Real-time Web Audio synthesis; all game sounds including sensory test stimuli are generated at runtime with zero audio files shipped.
- **@dnd-kit:** Pointer and touch drag-and-drop for sorting tasks.
- **Recharts:** Dashboard charting library.
- **Zustand v5:** Persisted client-side session state.

- **Resend:** Transactional email delivery for contact form notifications.
- **Vercel Analytics:** Zero-configuration visitor analytics.

Out of scope are clinical diagnosis, longitudinal multi-session tracking, medical-device certification and durable personal identity records.

## 1.5 Unfamiliarity of the Problem/Topic/Solution

Several serious game systems for ASD appear in the research literature, yet none currently combines all of the following characteristics in a single publicly available implementation:

- Browser delivery requiring no installation, covering the full 3 to 10 age range.
- Multi-domain coverage of all four ADOS-2 and DSM-5 domains within one session.
- Penalty-based scoring with multiplicative red-flag weighting grounded in three independent research papers.
- Real-time audio synthesis via Tone.js with no shipped audio files, enabling sensory-intensity gating without static asset overhead.
- HMAC-signed caregiver report links shareable by caregivers without requiring a researcher account.
- A researcher dashboard with pointer heatmap visualisation and Better Auth role-based access control.
- A production-ready open-source implementation backed by 312 automated tests and deployable on Vercel.

The novelty is inherent to the whole system rather than any one part; it's a research-based, multi-dimensional score system incorporated into a contemporary web application for commercial use, utilizing nothing but a serverless database and a free email server. Table 1.1 illustrates the connection between Horizons and relevant previous studies.

Table 1.1: Comparison of Horizons with related serious game screening tools

| **Feature** | **EmoGalaxy [4]** | **DTT Game [6]** | **ADOS-2 [3]** | **Horizons** |
|---|---|---|---|---|
| Browser-based | No | No | No | Yes |
| Multi-domain | No | No | Yes | Yes |
| Age range 3–10 | 6–14 | ASD children | 2–16 | 3–10 |
| Automated scoring | Yes | Yes | No | Yes |
| Red-flag detection | No | No | Yes | Yes |
| Pointer heatmap | No | No | No | Yes |
| Caregiver report | No | No | No | Yes |
| Open source | No | No | No | Yes |

## 1.6 Project Planning and Work Distribution

### 1.6.1 Project Planning: Gantt Chart

Figure 1.1 provides a visual timeline of the fifteen development phases across the sixteen-week schedule.

### 1.6.2 Work Distribution

- **Md. Shifat Hasan (Roll: 2107067):** System architecture design, PostgreSQL schema and Neon-hosted migration scripts, Zustand store architecture, penalty-based scoring engine across all domains and red flags, Chapter 1 gameplay (baseline name-response timing and joint-attention mechanics), Chapter 6 Grand Finale (cross-chapter consistency sampling and re-scoring), all game and researcher-dashboard API routes, Better Auth integration with role-based access control, researcher dashboard (charts, heatmaps and export), HMAC-SHA256 signed caregiver report generation, Tone.js audio synthesis module, Resend transactional email integration, full Bun test suite covering unit, integration and end-to-end scoring pipeline tests and project documentation.
- **Abu Hasanat Soykot (Roll: 2107100):** Chapter 2 to 5 gameplay implementations (Feeling World, Social World, Routine and Patterns and Pretend and Senses), drag-

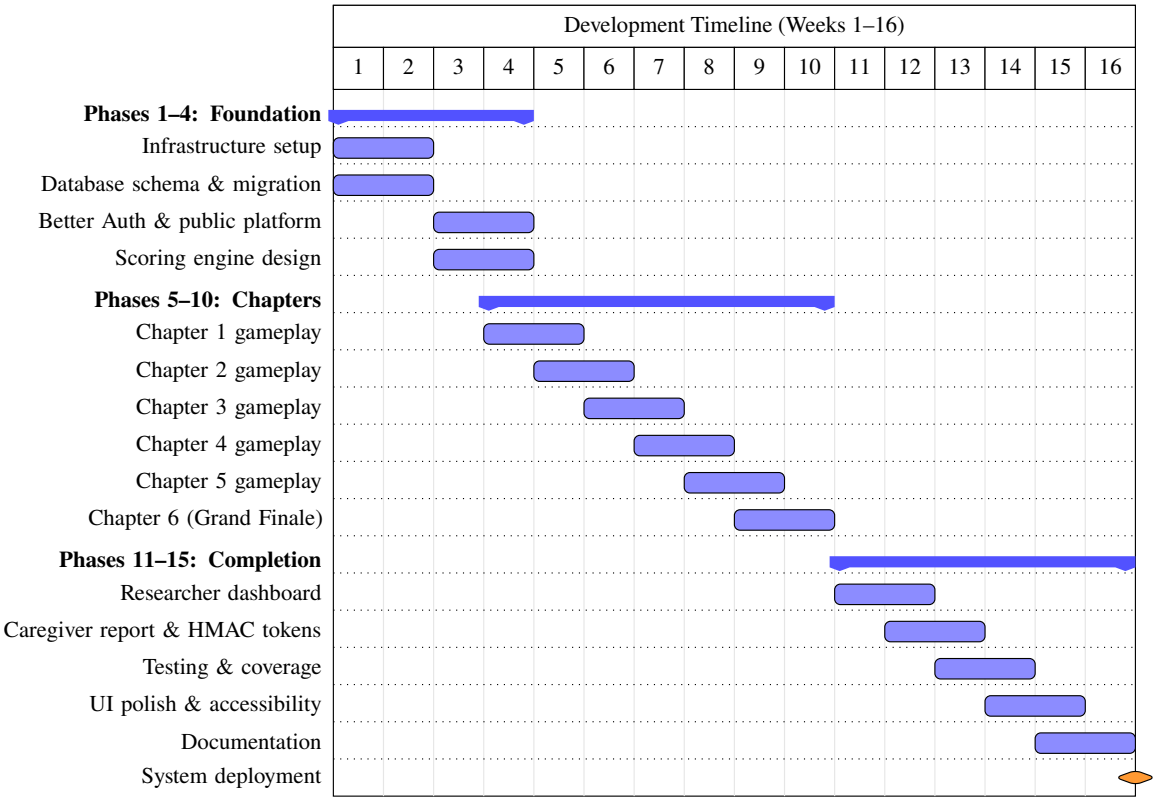


Figure 1.1: Project Gantt chart showing all fifteen development phases across sixteen weeks

and-drop sortable component using @dnd-kit, Framer Motion animation variants and sensory-intensity gating, emoji avatar customisation system, all public platform pages (landing, about, contact, survey) and mobile-responsive layout across the full application.

## 1.7 Applications of the Work

- **Clinical pre-referral screening:** Healthcare workers in resource-limited settings can deploy Horizons on any browser-capable device to gather pre-consultation screening data and the HMAC-signed caregiver report can be forwarded directly to the specialist without additional infrastructure.

- **Educational and developmental research:** Researchers studying developmental patterns can use the dashboard export function to analyse timestamped multi-domain behavioural data from complete sessions at scale.

- **Parental awareness:** Caregivers can review domain-level risk summaries through the signed report link, facilitating more informed referral conversations with their family physician.
- **Remote and telehealth screening:** The browser-based, zero-install nature of the system makes it suitable for telehealth programmes where a face-to-face clinical assessment is not immediately possible.

## 1.8 Organisation of the Project

The remainder of this report is structured as follows. Chapter II reviews related work in ASD screening, serious games and clinical assessment instruments. Chapter III describes the system methodology, including architecture, scoring design and the overall development approach. Chapter IV presents the implementation details, experimental setup, evaluation metrics and results. Chapter V addresses societal, health, ethical, legal and environmental considerations. Chapter VI maps the project to complex engineering problems and activities. Chapter VII concludes with a summary, identified limitations and directions for future work.

## 1.9 Conclusion

Chapter I has established the motivation, context and scope of the Horizons project. The gap between the clinical need for accessible ASD screening and the practical limitations of existing instruments was identified, six measurable objectives were articulated and the technology stack and multi-domain research foundation were outlined. The chapters that follow build systematically on this foundation.

# CHAPTER II

# Related Works

## 2.1 Introduction

This chapter reviews the peer-reviewed literature that directly shaped the design of Horizons, covering: clinical ASD assessment instruments; serious-game screening systems; and data-driven approaches to behavioural scoring. The review establishes both the research gaps Horizons addresses and the evidence base for its design decisions.

## 2.2 Related Works

### 2.2.1 ADOS-2 and the Limits of Social Communication as a Sole Predictor

Maddox et al. [3] assessed the ability of the ADOS-2 Module 4 in determining the validity of its diagnosis using an adult sample characterized by clinical complexity and recruited from community mental health clinics. One of the key findings was that this test showed perfect sensitivity by detecting all the individuals meeting the criteria for ASD. At the same time, 30% of individuals with psychosis had scores falling into the autism/ASD range according to the modified algorithm. The reasons for these differences were found in domain-level assessment: the ASD true-positive sample had virtually equal Social Communication scores (mean 10.5) as compared to the false-positive psychosis group. The ASD sample had much higher scores in another important domain – Restricted Interests and Repetitive Behaviours (mean score 4.7 compared to 2.0 among psychosis sample). The result is essential for calculating the Horizons score because the fact that Social Communication cannot differentiate between ASD and any other conditions means that 0.40 weight should be assigned to it.

### 2.2.2 DTT-Based Serious Game for ASD Vocabulary (Khowaja & Salim, 2018)

Khowaja and Salim [6] created a DTT-based serious game for vocabulary learning among children with ASD and tested the effectiveness of their approach through a single-subject withdrawal study that included baseline, intervention and maintenance periods. The research resulted in the identification of two criteria of evaluation that are implemented in Horizons in exactly the same manner: (i) task accuracy, measured as the ratio of correct answers to the overall number of answers given and (ii) average attempts per question, calculated as the total number of attempts made across the total number of questions asked. Furthermore, the research provided evidence for the importance of delineating phases of learning and practice from the phases of assessment, which is implemented in Horizons by way of providing practice-demofor every new mechanic type learned.

### 2.2.3 EmoGalaxy Emotion Screening Game (Irani et al., 2018)

Irani et al. [4] designed EmoGalaxy, a mobile emotion recognition game tested with 22 neurotypical children and 23 children with ASD across four emotion domains: happiness, sadness, fear and anger. Game scores were calculated as the ratio of correct answers to total moves. Applying four classification algorithms to the emotion-detection game data, support vector machine with a Gaussian kernel produced the best result at 93.3% accuracy. Kruskal-Wallis tests confirmed statistically significant group separation on both detection game scores (Sea: $p < 0.001$; Snowball: $p = 0.0007$), while face-scanner expression data did not reach significance, likely due to camera-quality variability. Three findings from this study directly shaped Horizons: the task score formula is adopted in Chapter 2 emotion tasks; the superior discriminative value of detection over expression tasks is reflected in the higher scoring weight given to emotion recognition; and the observation that children with ASD are less sensitive to negative emotions [12] motivated the red flag for negative-emotion recognition below 50%, which triggers a score multiplier of 1.20.

### 2.2.4 M-CHAT-R/F and Caregiver Screening Questionnaires

The Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F) [5] is a validated 20-item caregiver-report tool targeting children aged 16 to 30 months. Its design prioritises sensitivity: a total score of 8 to 20 on the initial checklist bypasses the follow-up interview and directs the child straight to diagnostic evaluation. Caregiver-report instruments depend on accurate parental recall and are therefore vulnerable to observer bias, but the behavioural indicators in the checklist are well validated and serve as a useful benchmark for game mechanic design. Several M-CHAT-R/F items map directly to Horizons tasks: response to name (Chapter 1, item 10), social smiling and reciprocity (Chapter 3, item 11), showing to share (Chapter 3, item 9), joint gaze following (Chapters 1 and 3, item 16) and imitation (Chapter 3, item 15).

### 2.2.5 Additional Related Systems

Daniels et al. [7] evaluated a wearable behavioural aid designed to support social learning in school-aged children with ASD, demonstrating feasibility for real-time social cue support outside the clinic. Wainer and Ingersoll [8] reviewed the broader evidence base for computer technology in teaching social communication to individuals with ASD, documenting consistent improvements in joint attention and imitation across multiple studies. Rudovic et al. [9] developed a personalised machine learning system for robot perception of affect and engagement in autism therapy sessions, illustrating how continuous behavioural monitoring can be made adaptive at the individual level. All three bodies of work are complementary to Horizons but focus on therapeutic intervention rather than pre-referral screening.

## 2.3 Discussion

Table 2.1 summarises the five comparison systems across six dimensions that matter for ASD screening tool evaluation. Horizons is the only entry that simultaneously provides browser delivery, multi-domain automated scoring, red-flag detection and pointer heatmap data, while grounding each design decision in independently published evidence.

Table 2.1: Summary comparison of related ASD screening and intervention systems

| **System** | **Type** | **Domains** | **Auto scoring** | **Browser** | **Red flags** |
|---|---|---|---|---|---|
| ADOS-2 [3] | Clinical | SC, RRB | No | No | Yes |
| EmoGalaxy [4] | Mobile game | SC (emotion) | Yes | No | No |
| DTT Game [6] | Desktop app | Language | Yes | No | No |
| Daniels et al. [7] | Wearable | SC | No | No | No |
| Wainer & Ingersoll [8] | Desktop | SC | Yes | No | No |
| **Horizons** | **Browser game** | **All 4** | **Yes** | **Yes** | **Yes** |

The review confirms that no existing system addresses the complete set of design goals targeted by Horizons. The proposed solution is therefore novel and represents a meaningful contribution to the field of technology-assisted ASD screening.

Tables 2.2 and 2.3 distil the principal research gaps uncovered by this review and record the corresponding outcomes that Horizons produces as a direct response to each gap.

## 2.4 Conclusion

The survey of ADOS-2 [3], DTT-based game scoring [6], EmoGalaxy [4] and M-CHAT-R/F [5] confirms that no single prior system addresses the full set of design requirements identified in Chapter I. The reviewed frameworks supply both the scientific justification and the concrete scoring formulae adopted in the Horizons scoring engine, which is described in the methodology presented in Chapter III.

Table 2.2: Research gaps identified from the related-work review

| Gap | Description | Source(s) |
|---|---|---|
| Browser-based delivery | No existing system offers zero-install, browser-based ASD screening for children aged 3–10 | [3,4,6] |
| Multi-domain coverage | No game-based tool simultaneously addresses all four ADOS-2 and DSM-5 indicator domains within a single session | [3,4,6] |
| Calibrated age range 3–10 | Prior tools target either toddlers or older school-age children; none span the full 3–10 range | [4,5,6] |
| Automated red-flag detection | Game-based systems lack automatic detection of clinically significant behavioural patterns with severity weighting | [4,6] |
| Shareable caregiver report | No reviewed system generates an integrity-protected report accessible without a researcher account | [3,4,6] |
| Pointer trajectory analytics | No tool records and visualises pointer or touch movement as a secondary behavioural signal | [3–7] |
| Open-source deployment | No openly deployable, production-quality, multi-domain implementation is publicly available | [3–9] |

Table 2.3: Research outcomes: how Horizons addresses each identified gap

| **Outcome** | **Horizons Implementation** |
|---|---|
| Browser-based serious game | Next.js 15 application; zero installation required; compatible with any modern browser on tablet or laptop |
| Four-domain gameplay coverage | Six chapters mapping to social communication, restricted and repetitive behaviour, pretend play and sensory processing |
| Research-informed scoring engine | Penalty model combining ADOS-2 domain weights, the EmoGalaxy task accuracy formula and the DTT average-attempts metric |
| Automated red-flag detection | Five flag types with multiplicative severity weights; multiplier stack capped at 2.0× to prevent score inflation |
| HMAC-signed caregiver report | SHA-256 signed token generated at session end; the link is shareable without requiring a researcher account |
| Researcher analytics dashboard | Per-session domain radar charts, pointer heatmaps, red-flag summaries and JSON or CSV data export |
| Automated test coverage | 312-test suite covering scoring unit tests, database queries, API routes and the full scoring pipeline |

# CHAPTER III

# Methodology

## 3.1 Introduction

This chapter describes the methodology used to design, architect and implement Horizons. The approach covers: problem design and requirements analysis; system architecture definition; scoring engine design grounded in the three research frameworks established in Chapter II; and the implementation strategy using test-driven development across a Bun runtime stack. Key methodological decisions are summarised at the end of the chapter together with their justifications.

## 3.2 Detailed Methodology

### 3.2.1 Problem Design and Analysis

The design process began with a systematic mapping from published ASD behavioural indicators to implementable game mechanics. Five criteria were applied to every proposed mechanic: it must derive from validated clinical literature; a child aged 3 to 10 must be able to complete it using pointer or touch input; it must not reveal the assessment purpose to the child or caregiver; its scoring must be deterministic and reproducible; and it must produce a quantitative penalty score expressible within the multi-domain model.

The requirements can be subdivided into the functional category (FR-1 to FR-12: session generation, six game modules, calculation of the results, report creation for caregivers, dashboard for researchers) and the non-functional one.

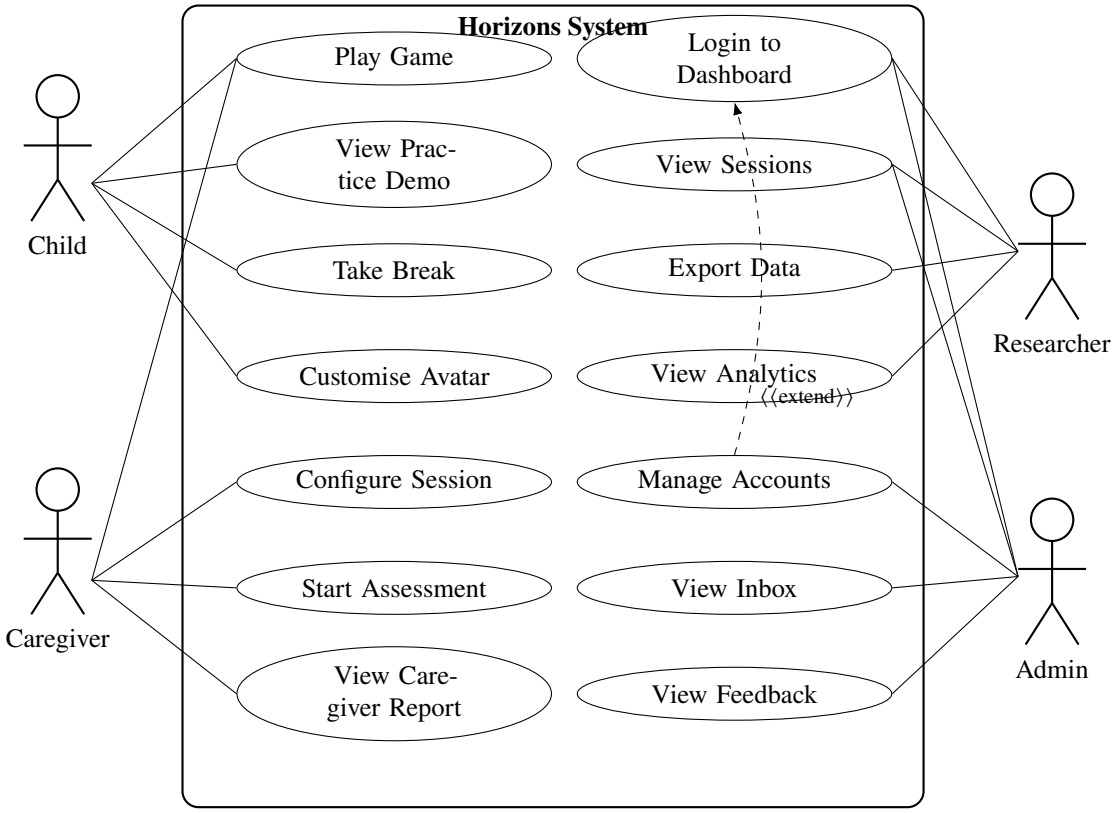


Figure 3.1: Use case diagram showing the four principal actors and their interactions with Horizons

Figure 3.1 maps each of the four principal actors to their respective system interactions. The Child actor engages exclusively with game-facing features, whereas the Caregiver configures the session and later accesses the signed report. Researcher and Admin accounts access progressively deeper data layers through the authenticated dashboard, with Manage Accounts, View Inbox and View Feedback restricted to the Admin role.

A game session goes through three distinct lifecycle states from creation till termination. The transition states are depicted in Figure. 3.2. A game session that has started moving through completion of its chapters will go through six different chapters until it reaches the *completed* terminal state, while if the user abandons the process then the terminal state becomes *abandoned*.

### 3.2.2 Overall Framework and Flowchart

Figure 3.3 shows the structure of the system where the child plays games provided by stimulus in the browser, events being sent via API routes into the database hosted on Neon using

PostgreSQL. In the end of session, scores engine generates an aggregate score, red flag multipliers and total risk score. An HMAC-SHA256 signature of report token is saved for each session and presented to caregiver. Researcher can access data for any session after logging into their own account via dashboard secured by Better Auth. Platform pages do not depend on game logic and authentication is not required there.

Figure 3.4 is a context-level DFD where the system is considered as one process while all external entities and dataflows passing the boundary are listed. Figure 3.5, level 1, describes six processes inside the main process and four databases used.

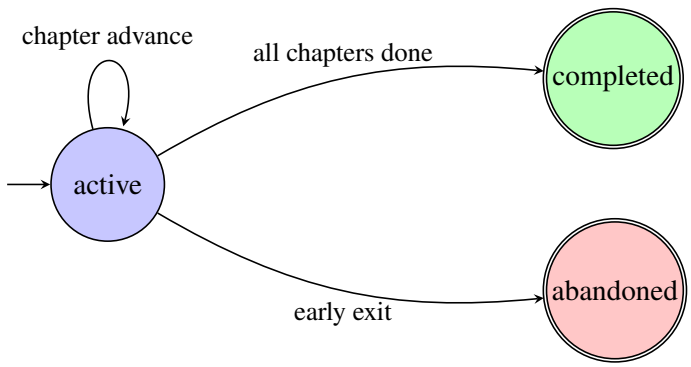


Figure 3.2: Session state transition diagram: three states with two terminal accepting states

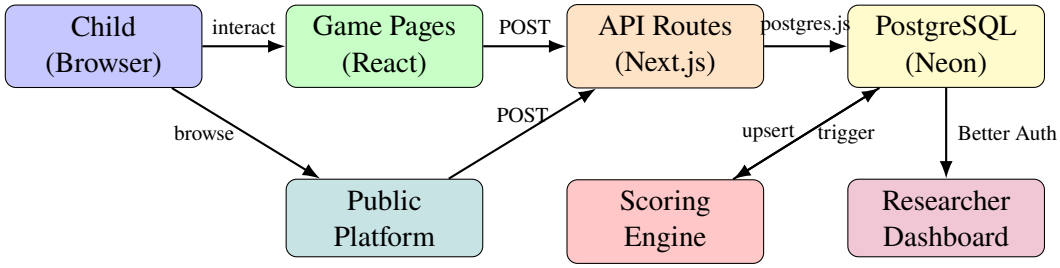


Figure 3.3: High-level system framework of Horizons

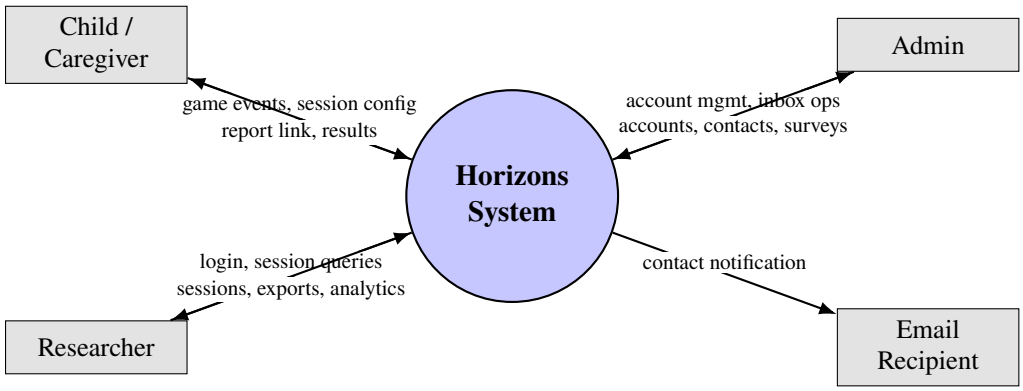


Figure 3.4: Data Flow Diagram Level 0 (context diagram) for the Horizons system

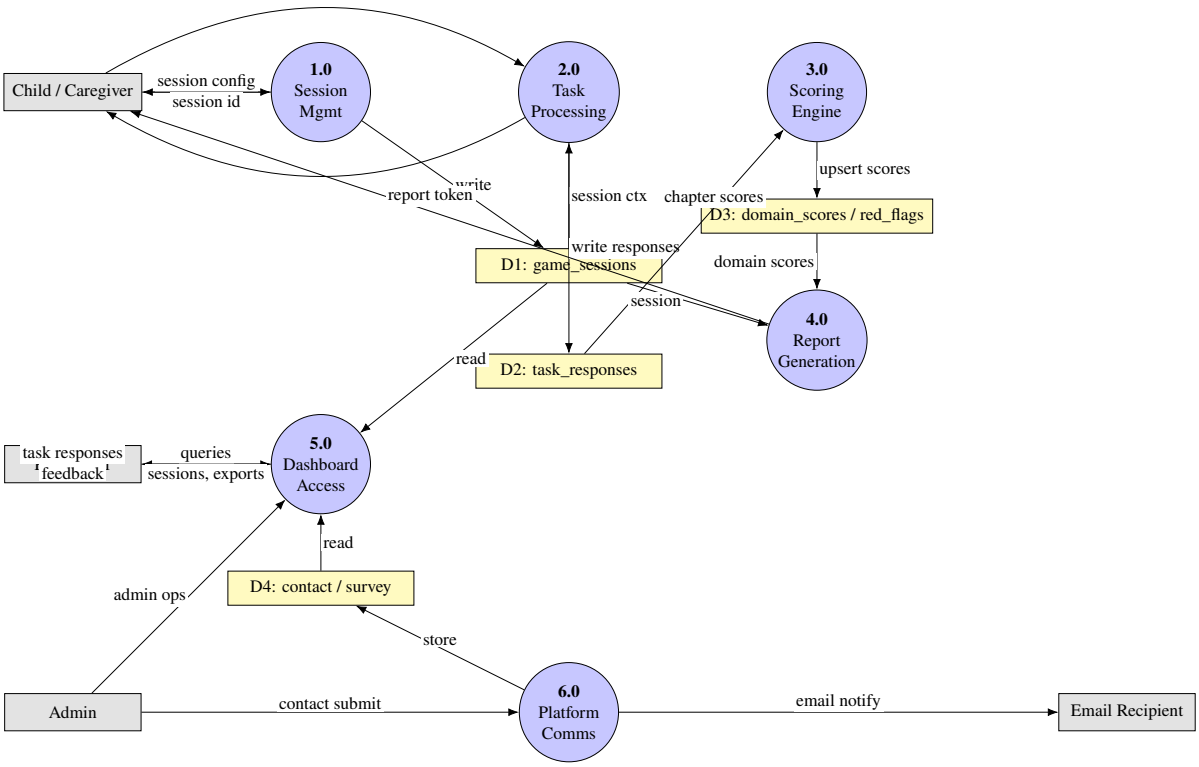


Figure 3.5: Data Flow Diagram Level 1: six sub-processes and four data stores

### 3.2.3 System Architecture

Horizons follows a three-tier architecture: browser (React game and platform pages with Zustand persisted state), application server (Next.js API route handlers with scoring logic and Better Auth session middleware) and data tier (PostgreSQL via Neon using the `postgres.js` driver). A critical constraint is that the database driver is never imported in client components; all persistence flows through server-side API routes, ensuring clean separation and maintaining test isolation.

The database schema comprises eight application tables: `game_sessions`, `task_responses`, `mouse_movements`, `chapter_scores`, `red_flags`, `domain_scores`, `contact_submissions` and `survey_responses`. Better Auth automatically provisions four additional tables (`user`, `session`, `account` and `verification`) for researcher authentication. Figure 3.6 illustrates the entity relationships among the primary game tables.

Figure 3.7 gives the parent overview of the four architectural tiers. Each tier is then elaborated individually in Figures 3.8 through 3.10.

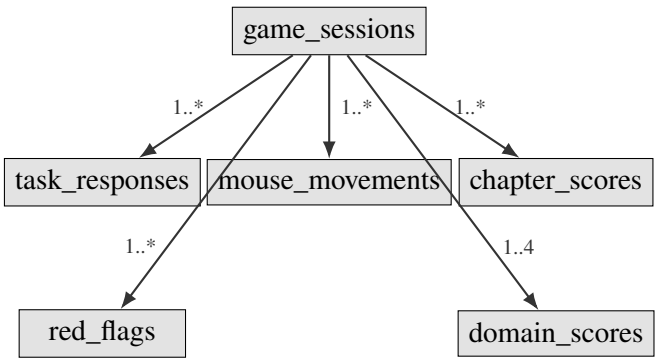


Figure 3.6: Entity relationship diagram for primary database tables

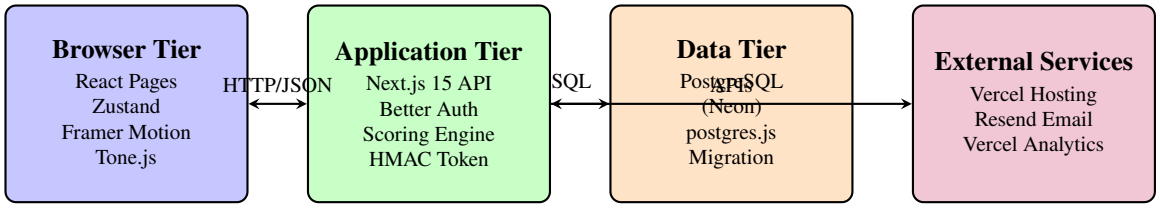


Figure 3.7: System architecture overview: four-tier parent block diagram

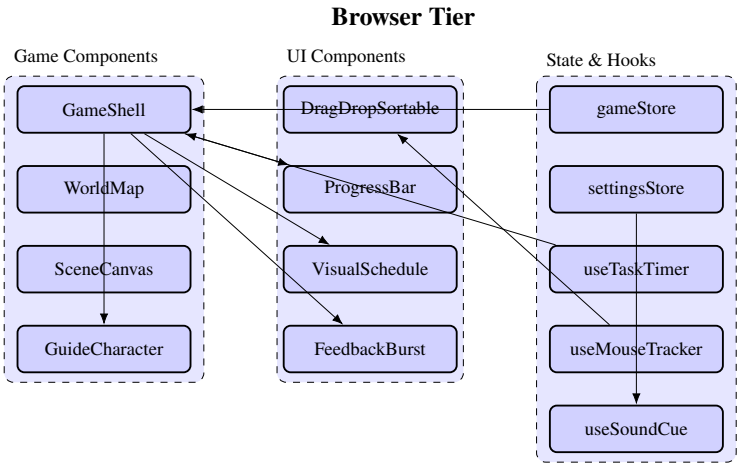


Figure 3.8: Browser Tier sub-diagram: game and UI components with state and hooks

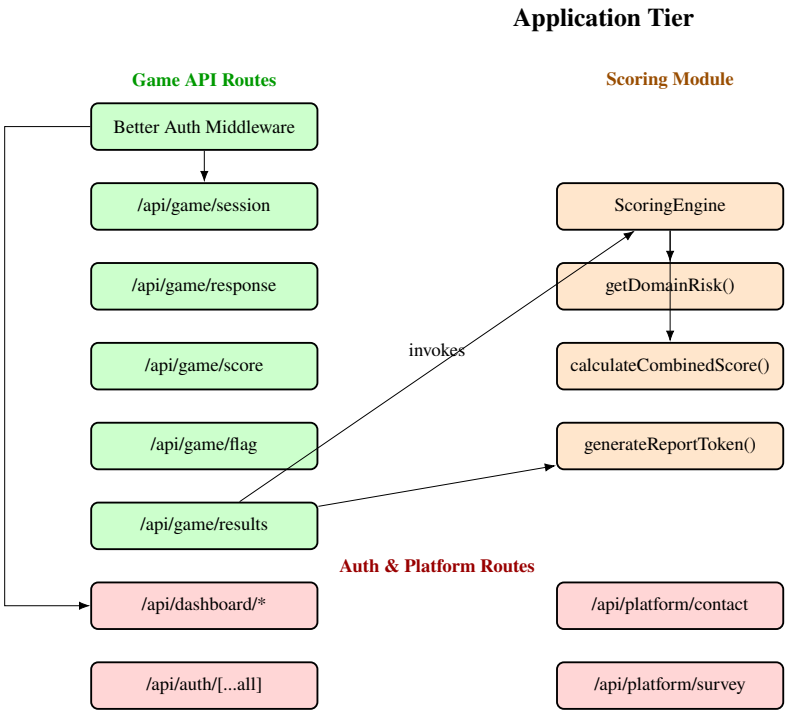


Figure 3.9: Application Tier sub-diagram: API route groups and the scoring module

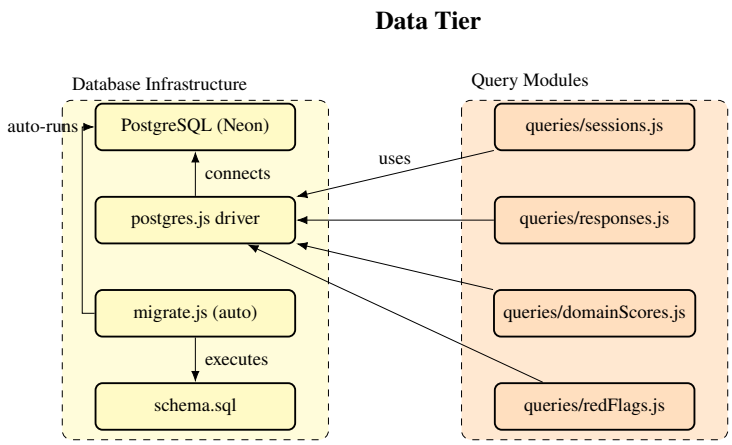


Figure 3.10: Data Tier sub-diagram: database infrastructure and query module organisation

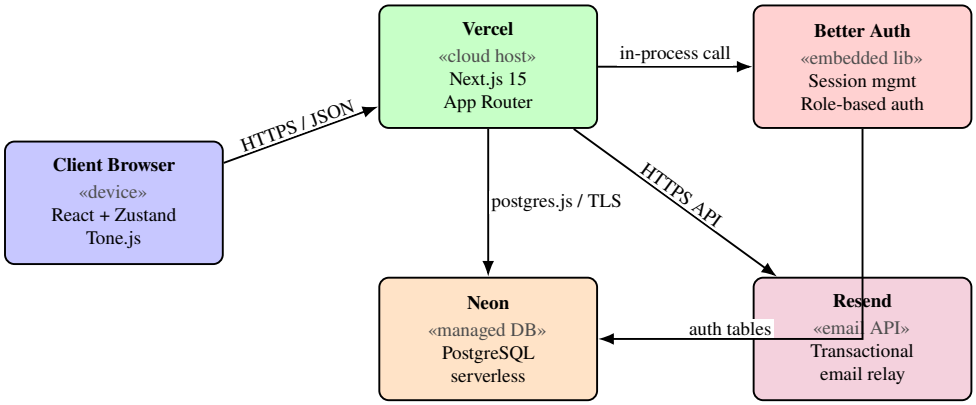


Figure 3.11: Deployment diagram: five nodes and their communication protocols

Figure 3.11 shows the five-node deployment diagram and the network protocols connecting them.

The full ER diagram in Figure 3.12 extends the simplified view in Figure 3.6 by showing all eight application tables and the key attributes of each.

Figure 3.13 presents the physical database schema showing each table's columns and data types.

Figure 3.14 presents the class diagram for the key server-side domain objects. `ScoringEngine` is implemented as a collection of pure functions (no mutable state) and is shown as a utility class; the remaining classes correspond directly to database table row objects managed through the query modules in `lib/db/queries/`.

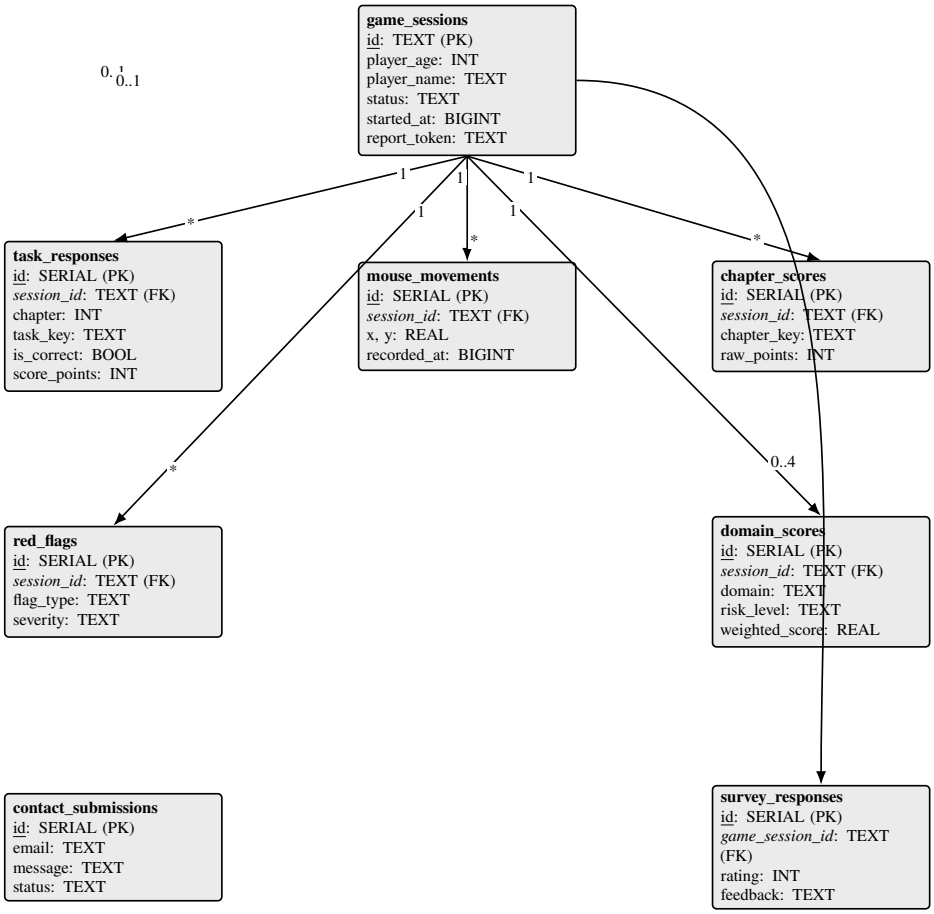


Figure 3.12: Full ER diagram: all eight application tables with key attributes and relationships

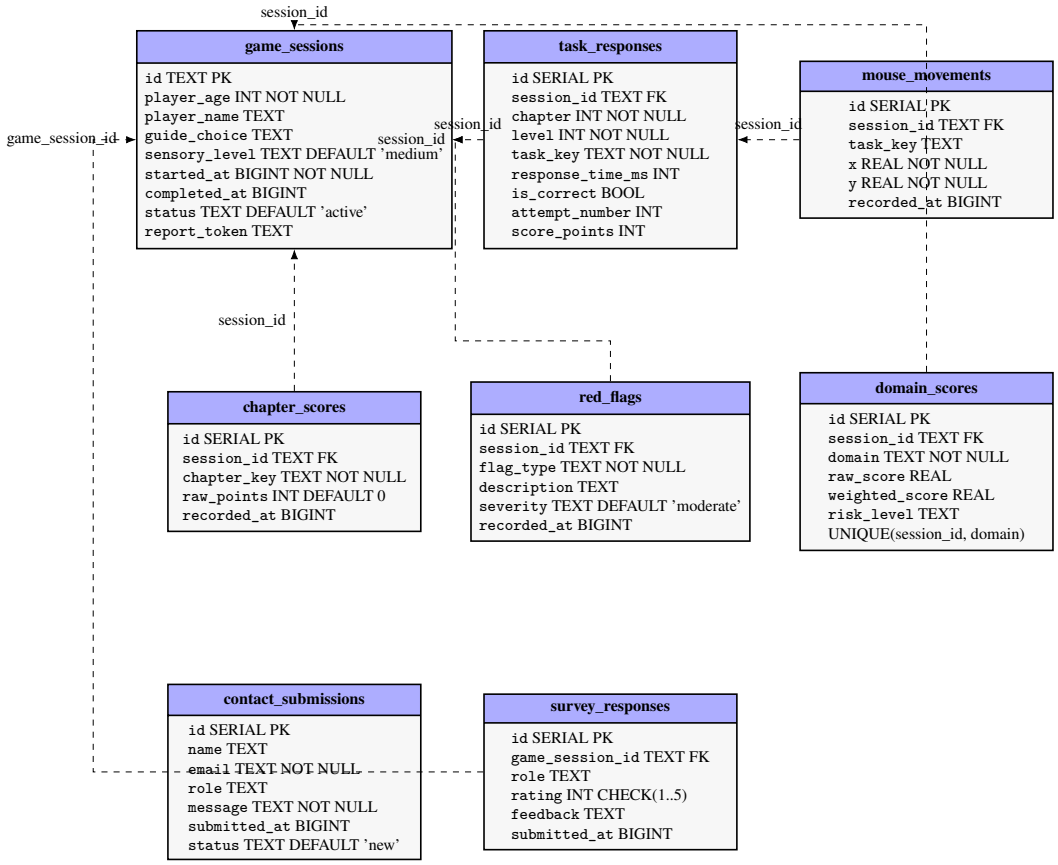


Figure 3.13: Database schema diagram: all eight application tables with columns, types and constraints

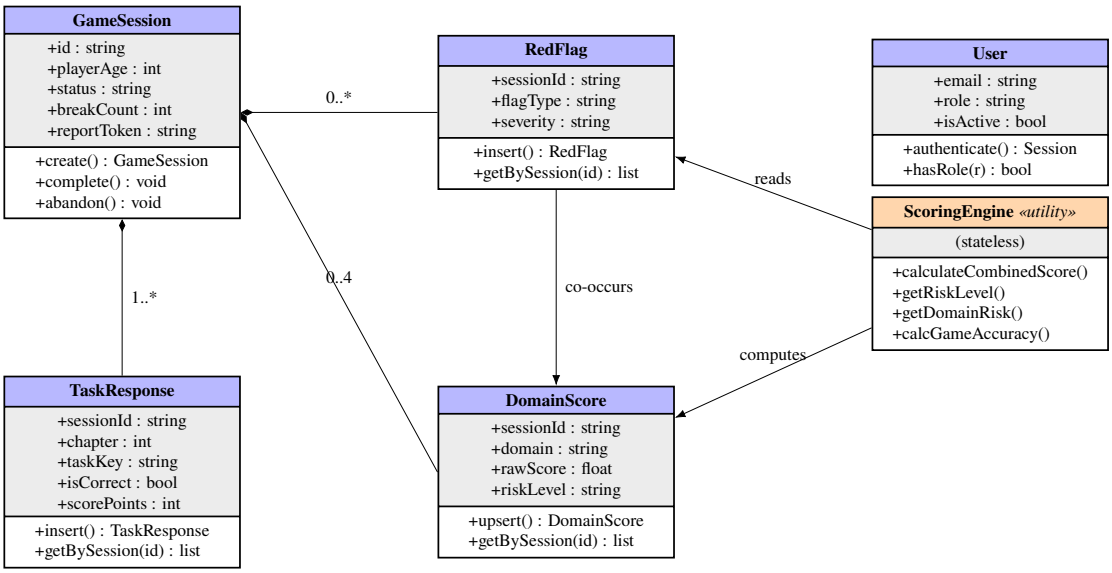


Figure 3.14: Class diagram for key server-side domain objects and the scoring utility

Table 3.1: Domain weights, maximum raw scores and risk thresholds

| **Domain** | **Chapters** | **Weight** | **Max score** | **High threshold** |
|---|---|---|---|---|
| Social communication | 2, 3 | 0.40 | 100 | ≥ 46 |
| Restricted/repetitive | 4 | 0.30 | 70 | ≥ 31 |
| Pretend play | 5 (Levels 1–2) | 0.15 | 40 | ≥ 21 |
| Sensory processing | 5 (Level 3) | 0.15 | 30 | ≥ 16 |
| **Combined** | 1–5 | 1.00 | - | ≥ 46 |

### 3.2.4 Scoring Engine Design

The scoring engine implements three research-grounded calculation modes.

**EmoGalaxy task accuracy (Irani et al., 2018):**

$$\text{score} = \frac{\text{correct_answers}}{\text{number_of_moves}} \tag{3.1}$$

**DTT average attempts (Khowaja & Salim, 2018):**

$$\bar{a} = \frac{\sum \text{attempts}}{\text{total_questions}} \tag{3.2}$$

**Combined weighted score with red-flag adjustment (ADOS-2 domain weighting):**

$$S_{\text{combined}} = \sum_{d} w_d \cdot R_d \times \min\left(\prod_{f \in F} m_f,\ 2.0\right) \tag{3.3}$$

where $w_d$ is the domain weight (Table 3.1), $R_d$ is the raw domain score, $F$ is the set of active red flags and $m_f$ is the multiplier for flag $f$. The combined multiplier is capped at 2.0× to prevent extreme score inflation from multiple co-occurring flags.

Five red flags are defined with multiplicative severity weights: negative emotion recognition below 50% (×1.20); complete absence of pretend play (×1.30); extreme sensory distress (×1.15); rigid pattern response plus distress at change (×1.20); and poor imitation across all modalities (×1.25). These are described in full in Appendix B.

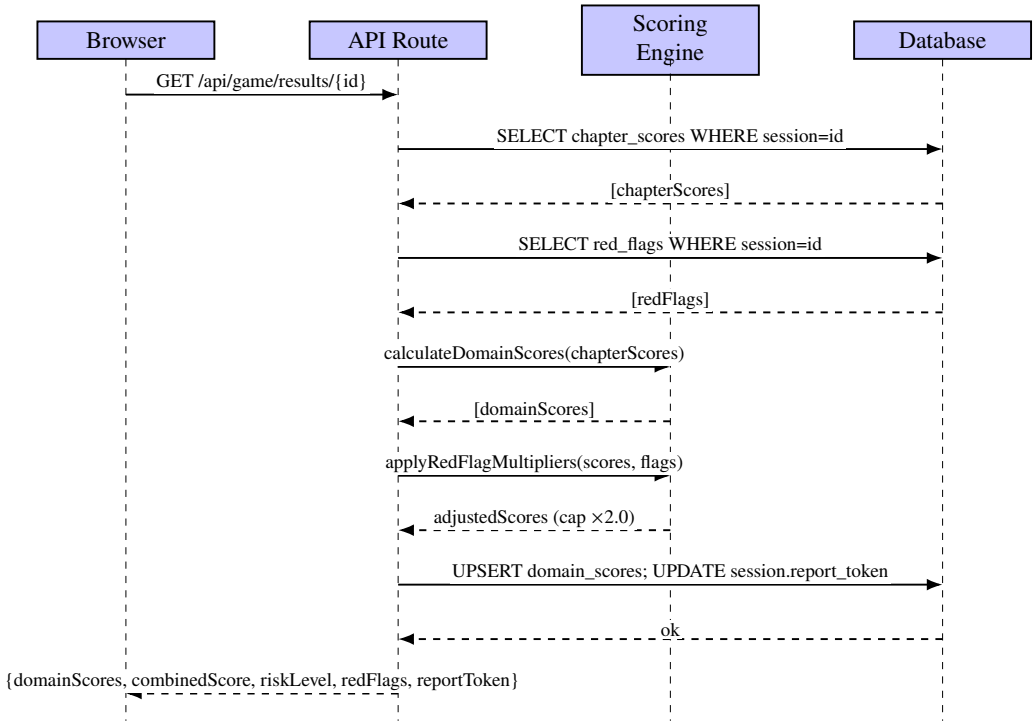


Figure 3.15: Sequence diagram: scoring pipeline from results request to complete result object

Figure 3.15 traces the message sequence that produces a completed scoring result from a single GET request to the results endpoint. The sequence is linear and involves four participants; no concurrent message paths are present.

### 3.2.5 Game Chapter Design

Each of the six chapters targets one or more ASD indicator domains through age-appropriate game tasks. The mechanics were derived by systematically mapping published ASD behavioural indicators to interactive scenarios that a child aged 3 to 10 can engage with through pointer or touch input alone. Table 3.2 lists those indicators alongside their clinical sources and the chapter level in which each is operationalised as a scored game task. Table 3.3 then summarises the broader chapter-to-domain structure.

Chapter 1 establishes a baseline through name-response timing and joint attention tasks but does not contribute to domain scoring. Chapters 2 and 3 together populate the social communication domain through three and four levels respectively; Chapter 3 Level 4 (Copy Cat) includes a false-belief Theory of Mind probe embedded in a conversation exchange. Chapter 4 addresses restricted and repetitive behaviour across four levels, with a drag-and-drop routine sequencing task that also measures cognitive flexibility when a planned disruption is introduced mid-sequence. Chapter 5 is divided between pretend play (Levels 1 and 2) and sensory processing (Level 3, the Sensory Explorer), where six Tone.js-synthesised sounds are presented one at a time and the child rates each using an emoji scale. Chapter 6 (Grand Finale) samples 15 tasks from Chapters 1 to 5 to check response consistency; it adds no new domain scoring.

Figure 3.16 shows the in-game activity flow from session start through to the results page. Decision nodes control three branches: whether the child requests a break during a task, whether additional scored tasks remain in the current level and whether further chapters remain after a chapter completion. The loop back from “Next Level” to “Chapter Introduction” reflects the level-by-level progression within each chapter.

Figure 3.17 traces the longer interaction chain across the full game session, from initial caregiver configuration through to report token delivery.

Three experience-management features run throughout the game. A sensory intensity toggle (low, medium and high), set by the caregiver before play begins, scales Framer Motion animation durations and Tone.js master volume. A visual schedule strip displayed between levels shows the child which tasks remain in the current chapter, reducing transition anxiety.

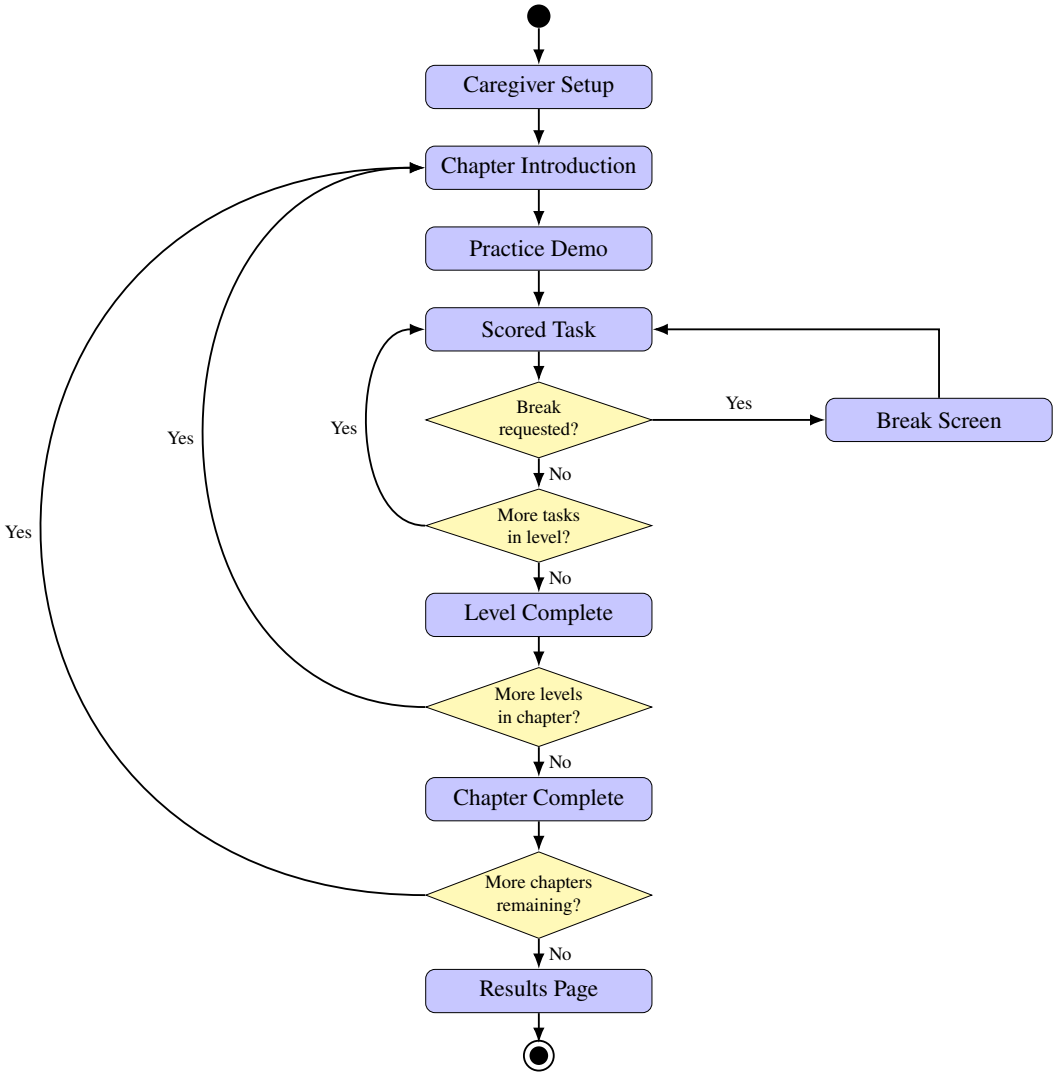


Figure 3.16: Activity diagram: in-game chapter flow from caregiver setup to the results page

A break button is always available; break frequency is logged as a behavioural data signal in the session record and is visible to researchers in the dashboard.

## 3.3 Conclusion

The methodology combines a systematic clinical-to-game mapping process, a three-tier web architecture with strict client and server separation and a research-grounded penalty scoring engine. Test-driven development with a 312-test automated suite ensures that every scoring function, database query and API route is independently verifiable. The design is intended to produce results that are reproducible, interpretable and directly comparable to the three

research frameworks cited in Chapter II.

Table 3.2: Suspected ASD behavioural indicators and their mapping to Horizons gameplay tasks

| **Behavioural Indicator** | **Clinical Source** | **Ch.** | **Level / Task** |
|---|---|---|---|
| Reduced or absent response to own name | M-CHAT-R/F item 10 | 1 | L1: Name Response |
| Joint attention deficit (gaze following) | M-CHAT-R/F items 9, 16 | 1 | L2: Guide Following |
| Difficulty recognising emotions in faces | ADOS-2 SC; EmoGalaxy [4] | 2 | L1: Emotion Matching |
| Impaired emotional regulation | ADOS-2 SC; DSM-5 | 2 | L3: Regulation Scenarios |
| Limited social smiling and reciprocity | M-CHAT-R/F item 11; ADOS-2 SC | 3 | L1: Greeting Sequence |
| Theory of Mind deficit (false-belief task) | ADOS-2 SC; DSM-5 A.1 | 3 | L2: Conversation & ToM |
| Reduced sharing and joint attention | M-CHAT-R/F items 9, 16; ADOS-2 | 3 | L3: Sharing & Attention |
| Imitation deficit across modalities | M-CHAT-R/F item 15; ADOS-2 | 3 | L4: Copy Cat |
| Insistence on sameness and rigid routines | ADOS-2 RRB; DSM-5 B.2 | 4 | L1–L2: Routine & Flexibility |
| Rigid adherence to repetitive patterns | ADOS-2 RRB; DSM-5 B.3 | 4 | L3: Pattern Detective |
| Narrow, restricted special interests | ADOS-2 RRB; DSM-5 B.3 | 4 | L4: Special Interests |
| Absence of symbolic or pretend play | M-CHAT-R/F item 3; ADOS-2 | 5 | L1–L2: Pretend Play |
| Atypical sensory reactivity (hyper- or hypo-) | DSM-5 B.4 | 5 | L3: Sensory Explorer |

Table 3.3: Chapter to domain mapping and task types

| Ch. | Title | Key Tasks | Domain |
|---|---|---|---|
| 1 | My World | Avatar customisation, response to name, joint attention following | Baseline |
| 2 | Feeling World | Emotion matching (drag-and-drop), expression mirror, regulation scenarios | SC |
| 3 | Social World | Greeting sequence, conversation and ToM probe, joint attention sharing, imitation | SC |
| 4 | Routine & Patterns | Morning routine sequencing, flexibility challenge, pattern detective, special interests | RRB |
| 5 | Pretend & Senses | Pretend play recognition, pretend world creation, sensory explorer | Pretend + Sensory |
| 6 | Grand Finale | 15 sampled review tasks, consistency check | - |

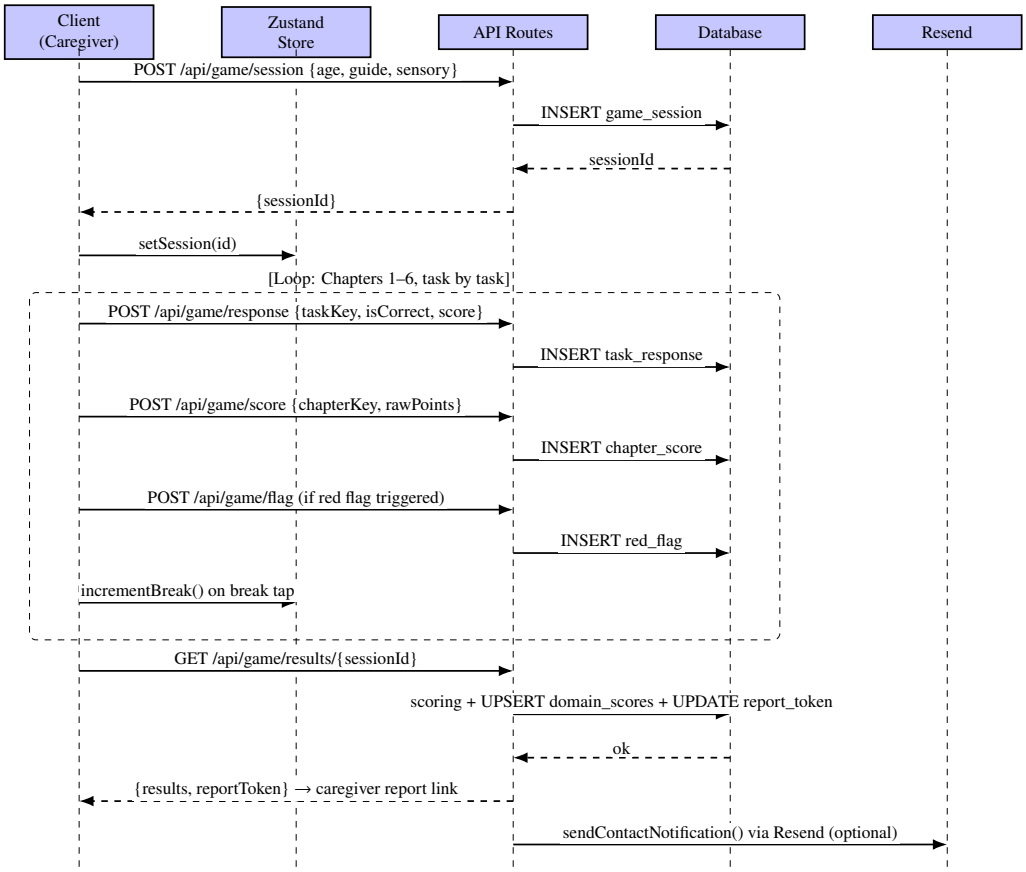


Figure 3.17: Sequence diagram: full game session flow from caregiver setup to signed report delivery

# CHAPTER IV

# Implementation, Results and Discussions

## 4.1 Introduction

This chapter covers the software implementation of Horizons.

## 4.2 Experimental Setup

The Horizons application was developed and tested on macOS using Bun 1.3.x. The PostgreSQL database was provisioned on a free-tier Neon project and accessed via the `postgres.js` driver with connection pooling. The production build was evaluated on a mid-range Android tablet (2 GB RAM, 1280×800 resolution) and on a standard laptop running Chrome 124.

Key system specifications are given in Table 4.1.

## 4.3 Implementation and Results

### 4.3.1 User Interface Wireframes

Figures 4.1 through 4.7 present the seven principal screen wireframes drawn to approximate proportions. Shaded regions indicate interactive or visually distinct zones within each screen.

## 4.4 Objective Achieved

All five objectives stated in Section 1.3 were achieved:

Table 4.1: Experimental setup and system specifications

| Parameter | Value |
|---|---|
| Framework | Next.js 15 (App Router) |
| Runtime | Bun 1.3.x |
| Database | PostgreSQL via Neon (`postgres.js` driver) |
| Authentication | Better Auth (email and password; researcher and admin roles) |
| Test runner | Bun built-in test runner |
| Total automated tests | 312 (all passing) |
| Primary viewport target | Tablet 768 px and above; mobile-first baseline 375 px |

1. Six gameplay chapters covering all four ADOS-2 and DSM-5 domains were implemented and deployed, with all 312 automated tests passing.

2. The penalty-based scoring engine incorporating EmoGalaxy, DTT and ADOS-2 formulae was implemented in `lib/scoring/` with complete unit test coverage.

3. Five red flags with multiplicative weighting were implemented and verified in integration tests using simulated full-session data.

4. The researcher dashboard with Better Auth authentication, domain charts, pointer heatmaps and JSON and CSV export was delivered and tested.

5. HMAC-SHA256 signed caregiver report tokens are generated at session completion; correctness is verified by both unit and integration tests.

## 4.5 Conclusion

The implementation of Horizons successfully delivers a production-standard, multi-domain ASD screening game. Software quality is supported by a comprehensive automated test suite. The system operates without locally hosted server infrastructure; the PostgreSQL database is serverless on Neon and the application deploys to Vercel, making it practical for research groups without dedicated server resources.

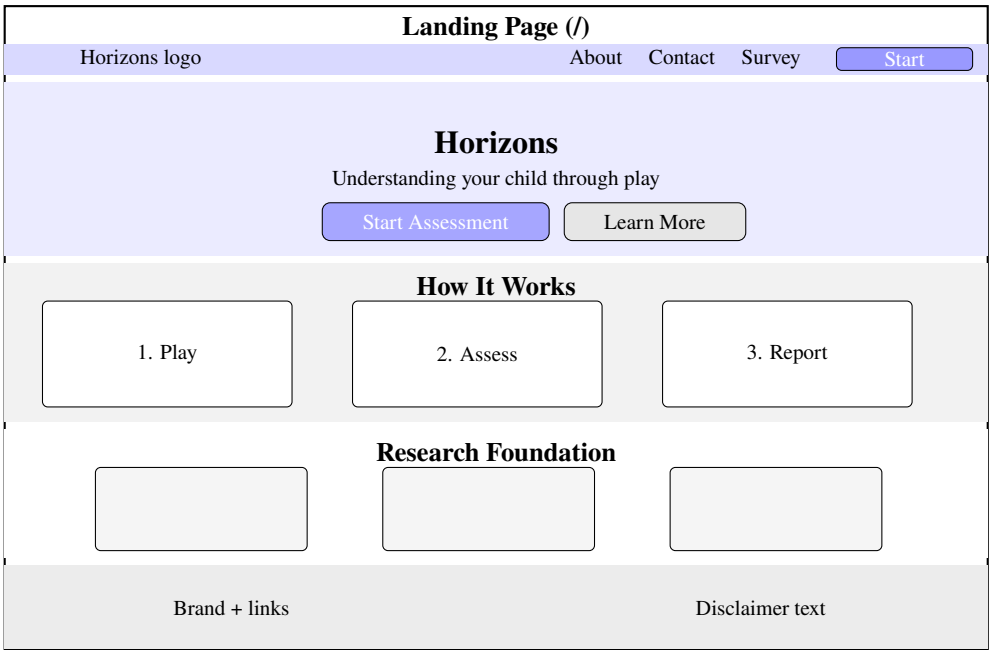


Figure 4.1: Wireframe: Landing page showing navigation bar, hero section, How It Works cards and footer

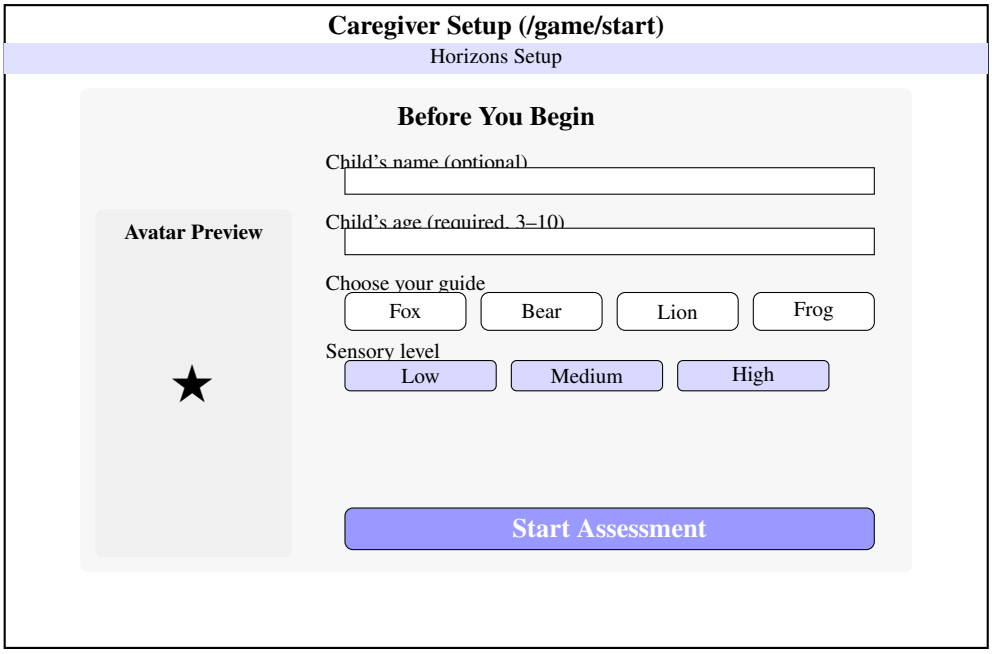


Figure 4.2: Wireframe: Caregiver setup screen showing form fields, guide selector and avatar preview

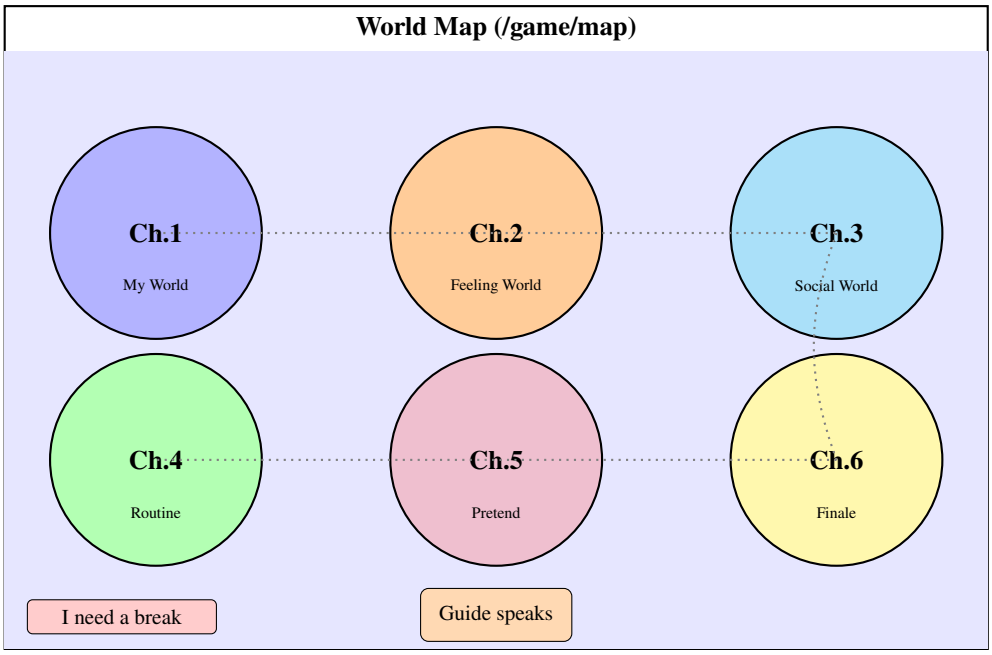


Figure 4.3: Wireframe: World Map showing six chapter islands with navigation path and break button

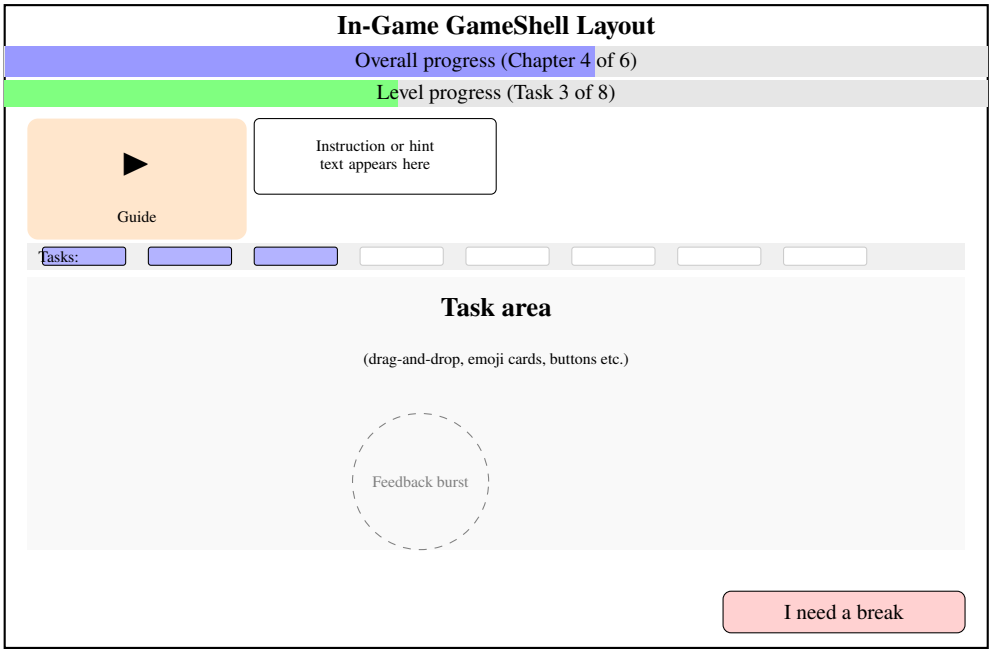


Figure 4.4: Wireframe: In-game GameShell showing dual progress bars, guide character and task area

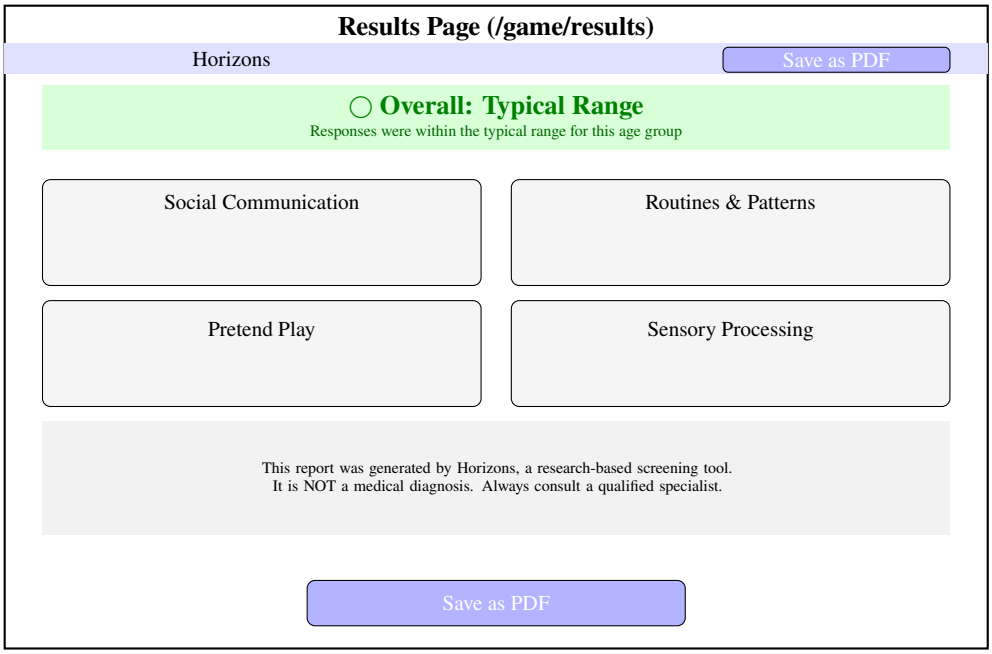


Figure 4.5: Wireframe: Results page with overall risk banner, four domain cards and disclaimer

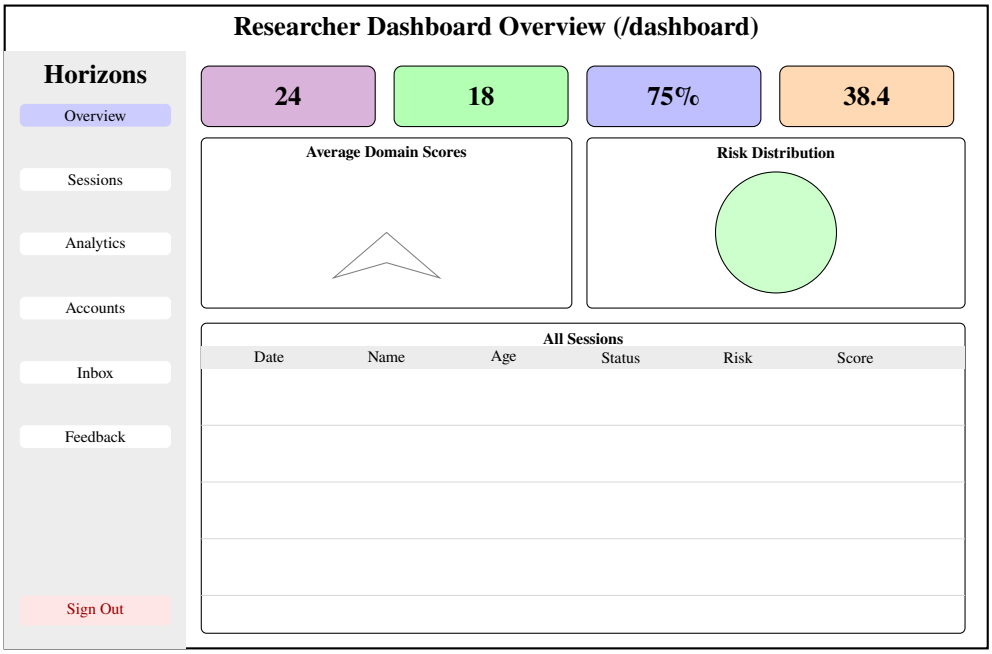


Figure 4.6: Wireframe: Researcher Dashboard overview with sidebar, stat cards, charts and session table

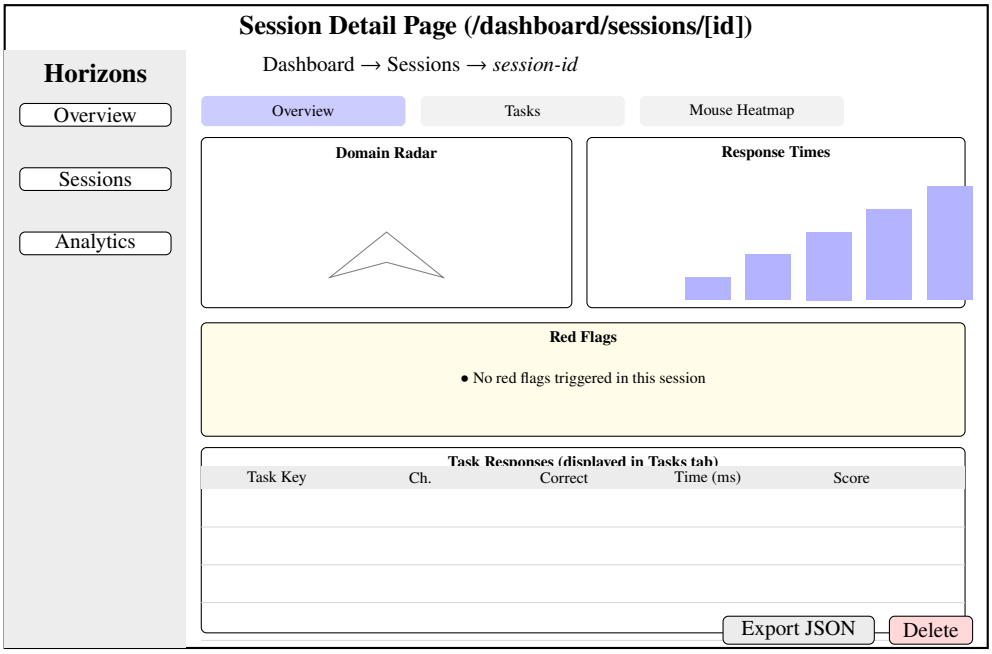


Figure 4.7: Wireframe: Session detail page showing tabs, domain radar, response time chart and task table

# CHAPTER V

# Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

## 5.1 Introduction

This chapter will take a look at some of the factors related to Horizons which go beyond mere technology. This includes issues such as intellectual property rights, ethics, safety concerns, legal obligations and other factors. All of these were considered right from the start when developing Horizons.

## 5.2 Intellectual Property Considerations

The Horizons software relies entirely on open-source libraries such as Next.js (MIT), Bun (MIT), Tailwind CSS (MIT), ShadCN UI (MIT), Framer Motion (MIT), Tone.js (MIT), @dnd-kit (MIT), Recharts (MIT), Zustand (MIT) and Better Auth (MIT). The Neon service is launched according to its free tier policy. As the program makes use of the PostgreSQL database with standard SQL commands using `postgres.js`, there is no vendor lock-in. All research instruments (ADOS-2, DTT game, EmoGalaxy) are mentioned only as research methodology references; no copyrighted assessment content is included in the program.

## 5.3 Ethical Considerations

The major ethical consideration concerning the ASD screening test for children concerns its possible misuse as a means for making a clinical diagnosis. The Horizons test eliminates this problem through several safeguards in its design process:

- An ongoing non-diagnostic warning is provided on the page showing the results and in the parent report: "Horizons is a screening guide only; it does not make a diagnosis. Please consult a qualified healthcare specialist for a proper evaluation."
- Domain risk levels are presented as categorical labels (low, medium, high, very high) rather than raw numerical scores on any child-facing or caregiver-facing surface.
- No photographs, video recordings, biometric data or precise location information are collected at any point.
- Player names entered at session start are optional display labels and are not retained after the session concludes.

## 5.4 Safety Considerations

Visual stimuli in Horizons were reviewed for photosensitivity risk. No high-frequency flicker patterns above 3 Hz are used in any Framer Motion animation, consistent with IEC 61966-2-2 guidance for digital displays. Tone.js audio output is attenuated in the low and medium sensory-intensity settings (approximately $-12$ dB and $-6$ dB relative to the high setting) to prevent auditory discomfort during the sensory test sounds in Chapter 5. All interactive game elements maintain a minimum touch-target size of 64×64 pixels and a minimum font size of 18 pt, reducing selection errors for young children who have not yet developed fine motor precision.

## 5.5 Legal Considerations

All session data is stored against UUID identifiers with no link to external identity records, consistent with the principle of data minimisation under GDPR and equivalent frameworks. Better Auth researcher sessions use HttpOnly, SameSite=Lax cookies with a seven-day renewing TTL; in-memory rate limiting enforces a maximum of five failed login attempts per fifteen-minute window, guarding against credential brute-forcing. Caregiver report links are signed using HMAC-SHA256 with a server-held secret stored in an environment variable; an unsigned or tampered token returns an error page rather than report data.

## 5.6 Impact of the Project on Societal, Health and Cultural Issues

Horizons has genuine potential to reduce the diagnostic gap in ASD by providing a free, browser-accessible screening pathway for families without convenient access to specialist clinics. The game framing removes much of the anxiety commonly associated with formal clinical assessment, which may encourage engagement in contexts where cultural stigma surrounds neurodevelopmental conditions. Because session data is stored anonymously and exportable in bulk, the researcher dashboard also supports population-level epidemiological studies in settings where ADOS-2 administration is logistically impractical.

## 5.7 Impact of the Project on the Environment and Sustainability

Horizons is a software product with no hardware manufacturing impact. The Neon serverless database platform runs on shared cloud infrastructure, which is substantially more energy-efficient per connection than a dedicated virtual machine running continuously for a single application. No image or audio files are distributed with the application; every sound is synthesised at runtime by Tone.js, which eliminates the bandwidth and storage footprint of a conventional asset library. Deployment on Vercel's shared infrastructure rather than a dedicated server further reduces the per-request energy cost. The zero-install browser-based delivery model also means that existing consumer devices such as tablets and laptops serve as the assessment platform, with no specialised clinical hardware procurement required.

## 5.8 Conclusion

Chapter V has reviewed the societal, health, environmental, safety, ethical and legal dimensions of the Horizons project. Strong privacy protections, explicit non-diagnostic disclaimers, open-source foundations, serverless infrastructure and zero-asset audio synthesis together ensure that the system is technically sound, responsibly designed and practical for deployment in clinical or educational contexts with minimal environmental and financial overhead.

# CHAPTER VI

# Addressing Complex Engineering Problems and Activities

## 6.1 Introduction

The chapter matches the Horizons Project with the criteria of complex engineering problem-solving and activities according to the Washington Accord and the Accreditation & Evaluation Committee for Engineering Education in Bangladesh (BAETE). This evidence shows that the project fulfills the complexity criteria of the graduate level according to the assessment model.

## 6.2 Complex Engineering Problems Associated with the Current Project

Horizons addresses several defining characteristics of complex engineering problems:

- **Depth of knowledge required:** The project demands simultaneous competence across clinical psychology (ASD assessment frameworks and M-CHAT-R/F behavioural indicators), full-stack software engineering (Next.js 15, PostgreSQL, React 19 and test-driven development), human-computer interaction design for children aged 3 to 10, real-time audio engineering (Tone.js Web Audio synthesis) and data privacy engineering.
- **Conflicting requirements:** The system must engage young children through visually rich, animated and playful interactions while collecting scientifically rigorous behavioural measurements that require precise timing, controlled stimuli and consistent task presentation. Game flow must never be interrupted by visible scoring logic, yet every interaction must be logged accurately without the child noticing.

- **Depth of analysis required:** Translating M-CHAT-R/F items and four ADOS-2 domains into implementable game mechanics required systematic analysis of clinical literature and iterative validation against three independent research frameworks. The selection of domain weights and red-flag multipliers was grounded in published evidence rather than developer intuition.

- **Involvement of multiple engineering disciplines:** The project spans full-stack web engineering, serverless database design, penalty scoring algorithm development, real-time audio synthesis, child-centred UX design and data privacy engineering. No single engineering specialism was sufficient.

- **Risk of misuse:** The clinical nature of the output creates a real risk of misinterpretation. Engineering controls such as the non-diagnostic disclaimer, categorical risk framing and HMAC-restricted caregiver report access were embedded in the system itself to mitigate this risk at the software level.

## 6.3 Complex Engineering Activities Associated with the Current Project

- **Design of a novel integrated system:** Horizons mandated the creation of an entirely new multidomain scoring system, as there were no precedents for such a system implemented before. Combining three research-driven systems into a penalty-driven system that multiplies the weight of red flags, with the use of a browser-based application without any pictures or sound elements but producing outputs via audio at runtime, is an innovative feat in engineering.

- **Managing competing stakeholders:** The application must satisfy three conflicting stakeholders at the same time: children should have an entertaining game with comprehensible content; parents must be able to interpret the results and be assured of the child's privacy and sharing of the report; and academics need the raw timestamps and statistical significance of results.

- **Use of modern and rapidly evolving tools:** Horizons uses the latest versions of Next.js 15, Bun 1.3, Tailwind CSS 4 and React 19, creating challenges in engineering under rapidly advancing technologies. Particular examples include asynchronous

page parameter feature introduced by Next.js 15, migration of Tailwind 4 to `@theme` CSS tokens instead of JavaScript configuration file and usage of connection pools provided by serverless Neon, which required additional research and learning beyond the official documentation.

- **System integration:** Horizons integrates several libraries into one Next.js app: drag-and-drop interactions provided by @dnd-kit, client-side audio synthesis by Tone.js (using lazy loading technique in order to comply with browser autoplay policy), physics-driven animations powered by Framer Motion (with sensory intensity gating), database interactions using serverless PostgreSQL provided by Neon and accessed via `postgres.js` library and role-based authentication through Better Auth.

## 6.4 Conclusion

Chapter VI has demonstrated that Horizons addresses the criteria for complex engineering problems and activities as defined by the Washington Accord and BAETE guidelines. The project required multi-disciplinary expertise, the resolution of genuinely conflicting design requirements and the integration of cutting-edge web technologies with clinical research frameworks. These characteristics confirm that the work represents a substantive engineering contribution appropriate for the System Development Project assessment.

# CHAPTER VII

# Conclusions

## 7.1 Summary

Horizons is a browser-based serious game for ASD screening that delivers six interconnected gameplay chapters covering social communication, restricted and repetitive behaviours, sensory processing and pretend play. The system is built on Next.js 15, Bun, PostgreSQL via Neon, Better Auth and Tone.js; it collects structured multi-domain behavioural data without installation or clinical equipment. No image or audio files are shipped with the application; all sounds are synthesised at runtime by Tone.js and all visual content is composed from emoji, CSS and SVG. A public-facing platform, a Better Auth secured researcher dashboard and an HMAC-SHA256 signed caregiver report link complete the system. A penalty-based scoring engine grounded in ADOS-2, DTT game design and EmoGalaxy computes weighted domain scores and a four-level combined risk classification. An automated test suite of 312 tests verifies the correctness of scoring, database and API logic.

## 7.2 Limitations

- **Cultural adaptation:** The social scenario content in Chapters 3 and 4 has not been formally validated for cultural neutrality across diverse populations; some scenarios may carry culturally specific expectations about greetings or daily routines.
- **No longitudinal tracking:** The current version does not record within-child changes across multiple sessions, which limits its utility for monitoring intervention outcomes over time.

## 7.3 Recommendations and Future Works

- **Large-scale clinical validation:** A multi-center study including 200 or more participants, with diverse age, culture and clinical subgroups, will make the task of threshold calibration and receiver operating characteristics curve easier.

- **Machine learning enhanced scoring:** The training of a support vector machine or gradient-boosted classifier on feature vectors constructed based on data collected throughout game play can outperform or at least equal the 93% accuracy of Irani et al.'s EmoGalaxy classifier, which includes all four categories.

- **Age-stratified scoring:** Developing separate scoring coefficients for the different age subgroups 3–5, 6–8 and 9–10 years will improve classification at extremes of development.

- **Longitudinal session tracking:** The inclusion of an opt-in session history will allow individual child progress monitoring when playing multiple times.

- **Multilingual and cultural adaptation:** Adaptation of stimulus text and social situations according to the particular culture will increase Horizons' usefulness in other languages beyond English.

- **Integration with referral workflows:** The inclusion of the HMAC signed report from the caregiver in the referral pathway into the clinic's patient management program will ensure that Horizons can be used as a pre-referral screener rather than a stand-alone assessment.

# Appendices

## Appendix A: API Route Summary

Horizons API route reference

| Method | Route | Description |
|---|---|---|
| POST | /api/game/session | Create new game session |
| GET | /api/game/session/[id] | Retrieve session record |
| PATCH | /api/game/session/[id] | Update chapter, level or status |
| POST | /api/game/response | Log a task response |
| POST | /api/game/score | Add chapter score points |
| POST | /api/game/mouse | Batch pointer movement log |
| POST | /api/game/flag | Add a red flag |
| GET | /api/game/results/[sessionId] | Compute and return full session results |
| GET | /api/game/chapter6/tasks | Return sampled Grand Finale task list |
| POST | /api/platform/contact | Submit contact form; triggers Resend email |
| POST | /api/platform/survey | Submit post-session survey |
| GET/POST | /api/auth/[...all] | Better Auth catch-all handler |
| GET | /api/dashboard/sessions | List all sessions (researcher) |
| GET | /api/dashboard/sessions/[id] | Session detail (researcher) |
| DELETE | /api/dashboard/sessions/[id] | Delete session record (researcher) |
| GET | /api/dashboard/export | JSON or CSV data export (researcher) |

(continued)

| Method | Route | Description |
|---|---|---|
| GET | /api/dashboard/accounts | List researcher accounts (admin) |
| POST | /api/dashboard/accounts | Create researcher account (admin) |
| PATCH | /api/dashboard/accounts/[id] | Update role or active status (admin) |
| GET | /api/dashboard/contact | List contact submissions (admin) |
| PATCH | /api/dashboard/contact/[id] | Update contact submission status (admin) |
| GET | /api/dashboard/survey | List survey responses (admin) |

## Appendix B: Scoring Red-Flag Multipliers

Red flag types, trigger conditions and score multipliers

| Flag Type | Condition | Multiplier |
|---|---|---|
| `negative_emotion recognition_under_50` | Negative-emotion accuracy < 50% in Ch. 2 Level 1 | ×1.20 |
| `complete_absence pretend_play` | All 5 pretend-play recognition trials answered literally in Ch. 5 Level 1 | ×1.30 |
| `extreme_sensory distress` | ≥4 of 6 synthesised sounds rated distressing in Ch. 5 Level 3 | ×1.15 |
| `rigid_pattern distress` | Meltdown at forced error plus insistence on returning to the original pattern in Ch. 4 Level 3 | ×1.20 |
| `poor_imitation all_modalities` | ≥5 errors across facial, body and object modalities in Ch. 3 Level 4 | ×1.25 |

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